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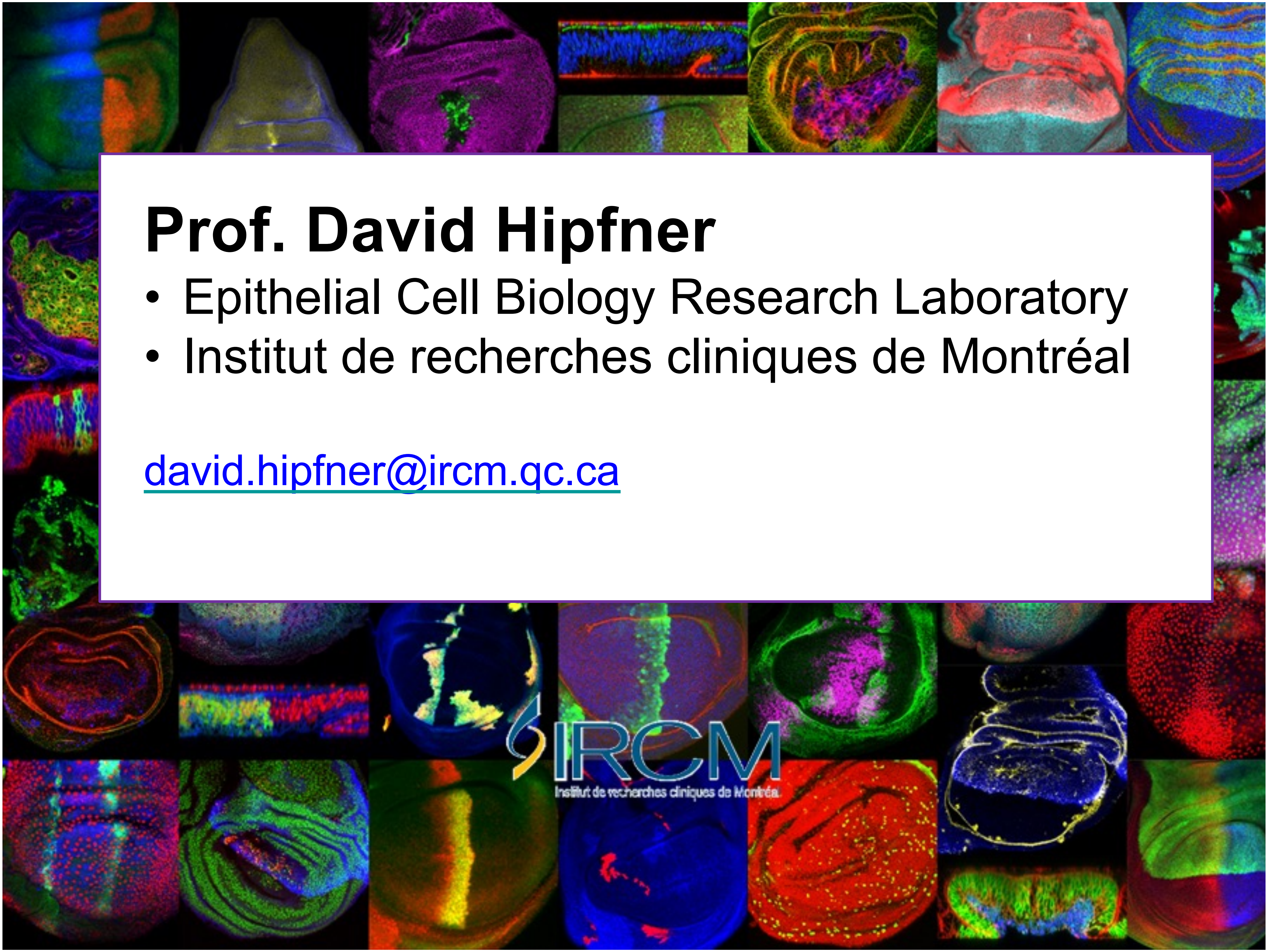

How to reach me

If you have questions about lecture materials or problem sets, you can get a hold of me:

1. **Online right after class** - I have nothing urgent planned
2. **Via the discussion board** - if you have a question, so do others. Post it there, I will let other students and/or TAs have first try and will answer within ~24 hours
3. **By email** – Please use my IRCM email address: **david.hipfner@ircm.qc.ca** . If you email questions I will post them (anonymously) and follow similar protocol as above

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Lecture Materials

ORGANIZATION

- Readings are posted in the slides at the start of each lecture (also in the lecture schedule document in MyCourses).

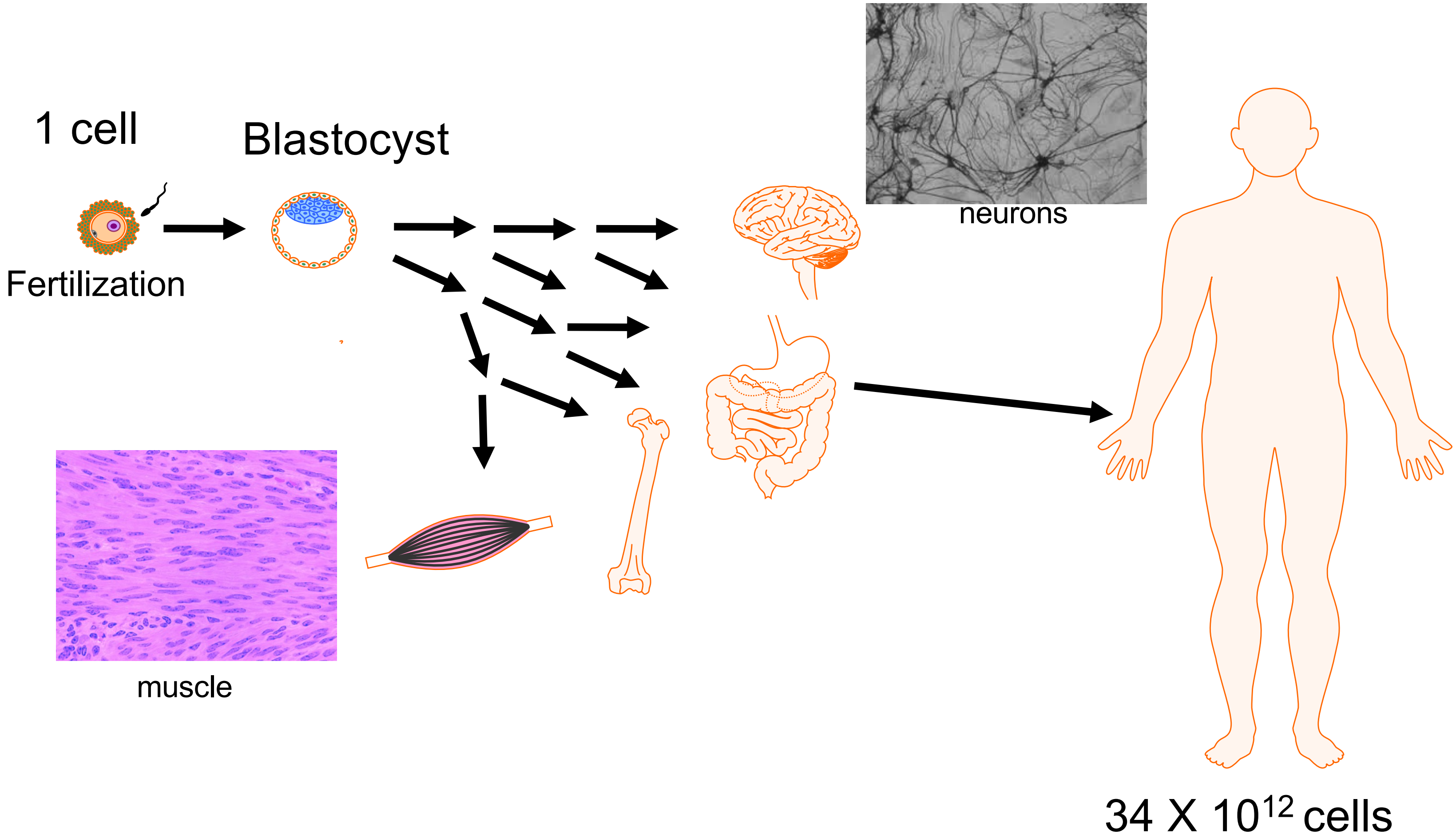
- The Primers include background material that most, but perhaps not all, students will be familiar with. Everyone is expected to read through them and to know the information they contain.

- Primers also contain problem sets – textbook questions for each lecture, plus some extra practice questions on material not well covered in the textbook.

- I will post answers for the problem sets (in the Weekly practice questions section of MyCourses), with a delay of a couple of days to give people a chance to do them.

- We'll fall behind schedule today, will catch it up at the end!

Why is transcription important?



Almost every cell in our body has essentially the same DNA sequence.

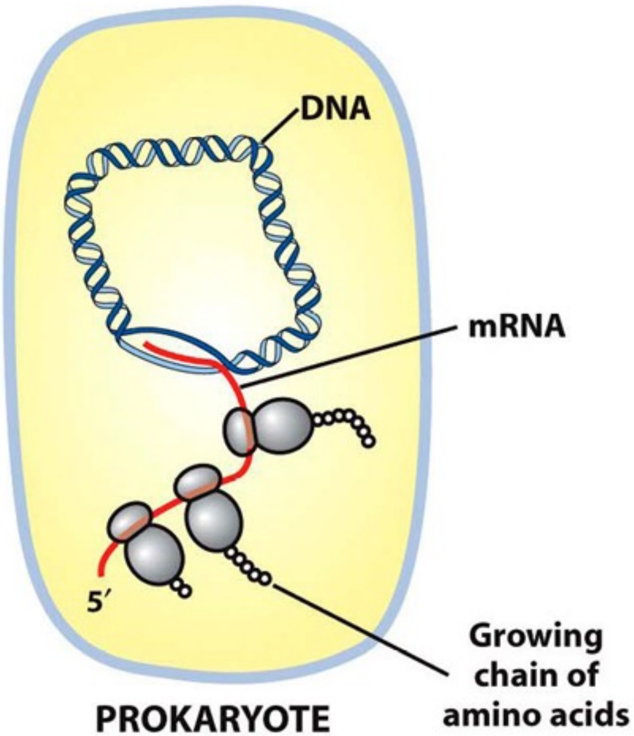
What makes each cell type unique is largely determined by the expression of specific subsets of tissue-specific genes (mostly controlled at the level of transcription).

Lecture 27: Genetics & Gene Regulation I –Transcriptional regulation in prokaryotes

Readings:
- Ch 8 271-276 [section - 8.2, not mRNA decay]
- Ch 11 369-383 [sections Intro - 11.3]

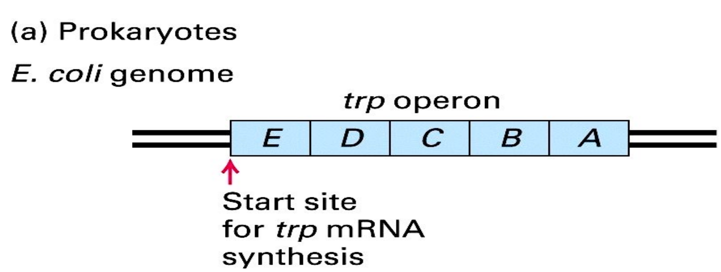
- Control of transcription initiation in prokaryotes
  - *cis* DNA sequences & *trans*-acting transcription factors
  - positive vs. negative regulation
  - example – *E. coli lac* operon

Regulating gene expression in prokaryotes



Simple system, relatively few levels at which “gene activity” can be regulated

Transcription initiation **(to transcribe or not…?) is one of the main mechanisms for regulating genes in prokaryotes**



Many prokaryotic metabolic genes involved in the same process are organized into **operons** (one transcript unit coding for multiple proteins under a single promoter)

Organization of prokaryotic genes

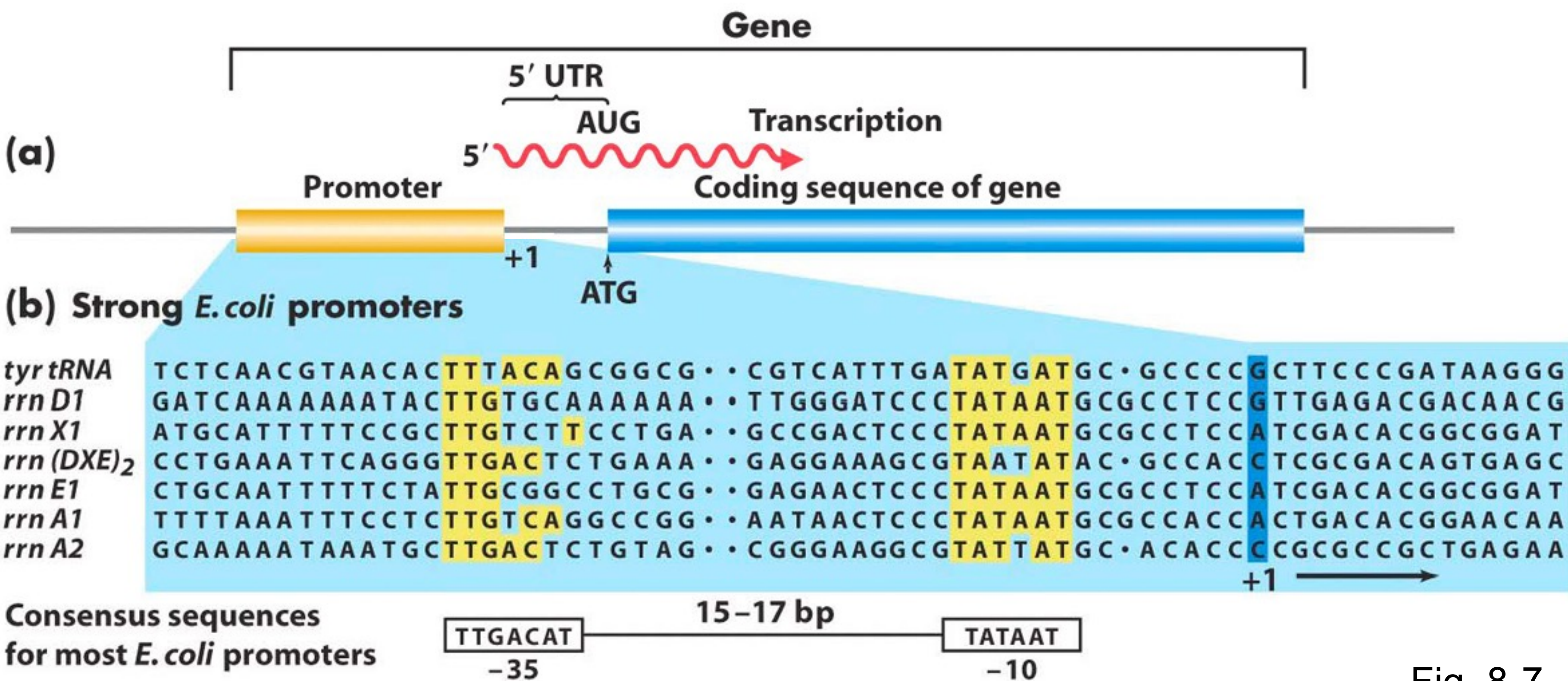


Fig. 8-7

RNA polymerase

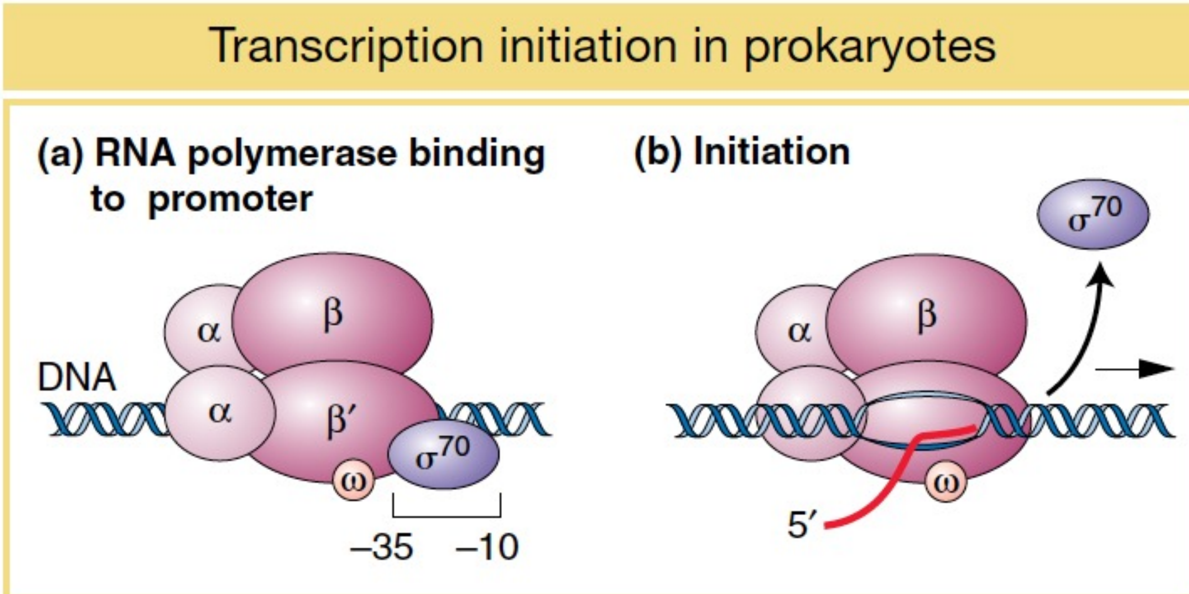


Fig. 8-8

- **Sigma factor subunit of RNA polymerase** binds to -35/-10 promoter sequences to properly position the holoenzyme at transcription start site
- In prokaryotes, RNA polymerase easily binds the promoter – default state of genes is **“ON” (a bit…)**

How is transcription regulated?

Regulating transcription: Activators and Repressors

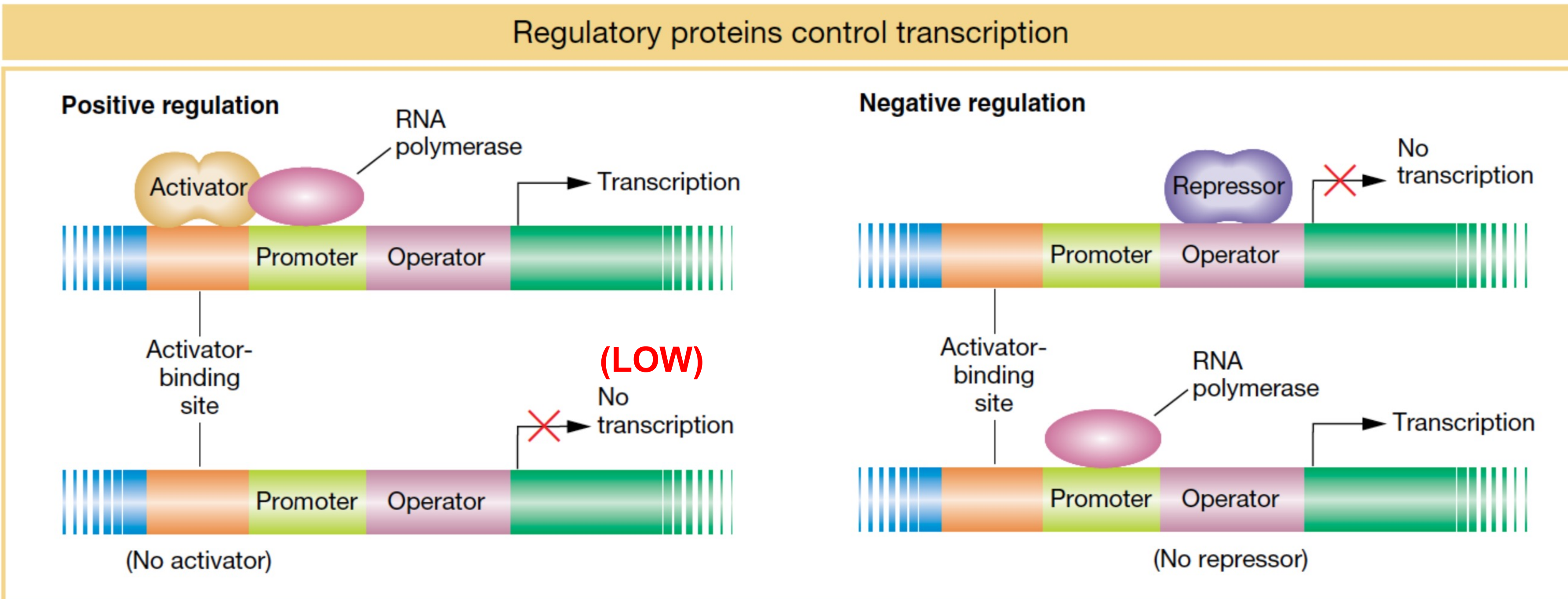
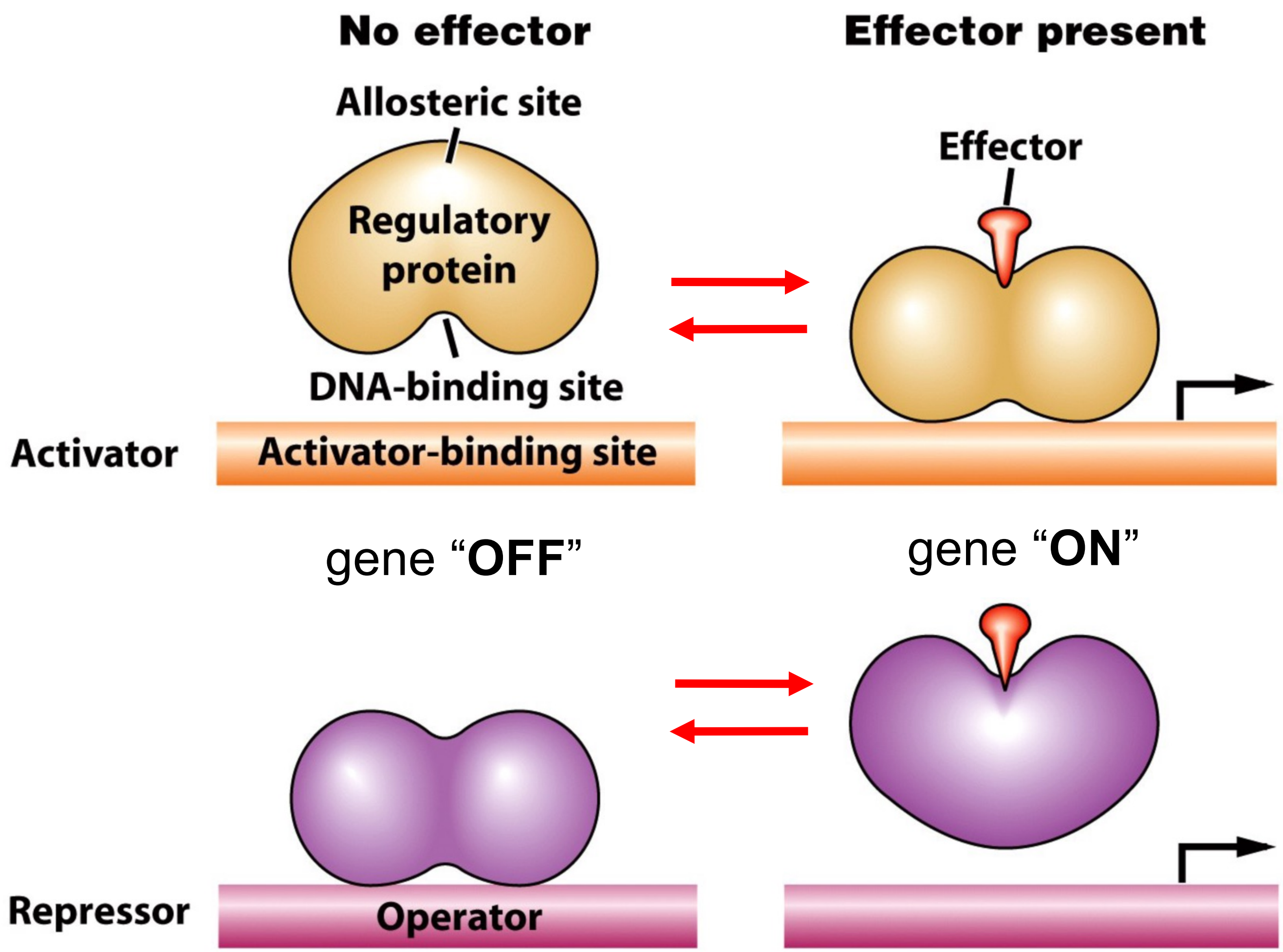


Fig.11-2

Binding of sequence-specific DNA binding proteins around the promoter either "**activates**" or "**represses**" gene transcription

(Operator = binding sites for repressors)

"Genetic sensors" - Allosteric regulation of transcriptional activators/repressors



"allosteric transition"

Fig.11-3

- Effector called an "inducer" if its presence leads to increased gene expression

In a strain of bacteria, addition of glutamine to the growth medium causes the expression of four genes involved in the glutamine biosynthetic pathway to each <u>decrease</u> 20-fold. Without knowing anything more about the system, which of the following statements is most consistent with this data?

A. the glutamine system is under positive regulation
B. the glutamine genes are arranged in an operon
C. glutamine is an inducer of the system
D. glutamine prevents a repressor from binding to the operator

Classical studies of gene regulation – The *lac* operon

Figure 11-1 François Jacob, Jacques Monod, and André Lwoff were awarded the 1965 Nobel Prize in Physiology or Medicine for their pioneering work on how gene expression is regulated. [*The Pasteur Institute*.]

- investigated regulation of the *lac* operon in *E. coli*, required for lactose metabolism
- Used a purely genetic approach to understand gene regulation, later confirmed by molecular techniques (e.g. biochemical characterization of repressor protein)

Components of the *Lac* operon

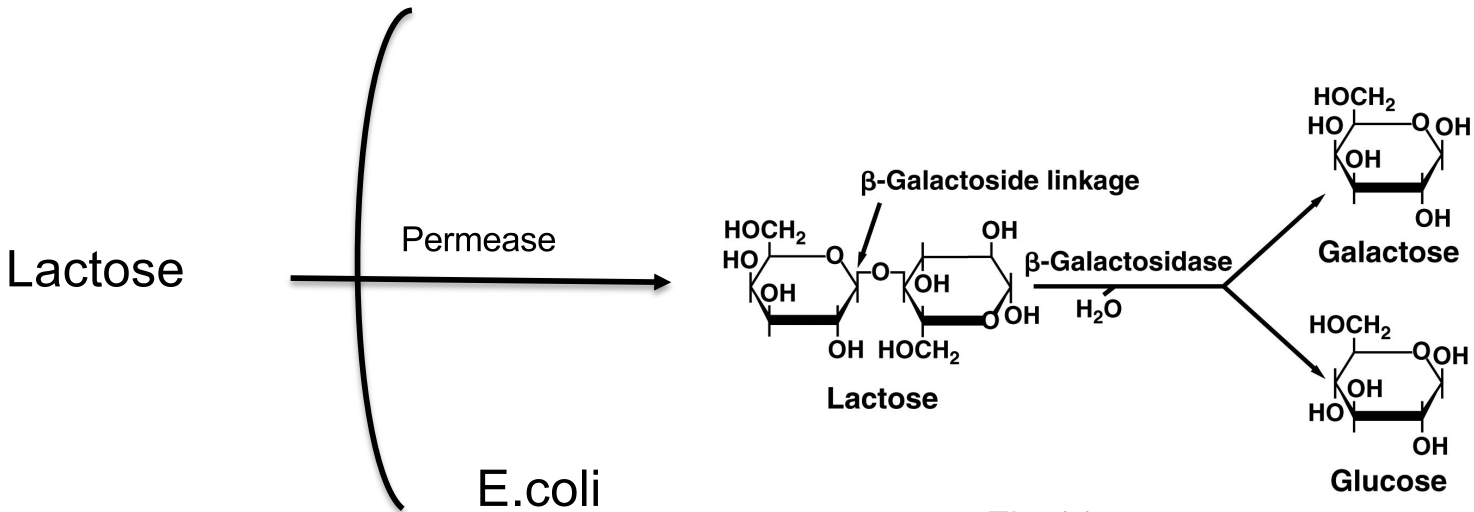


Fig.11-5

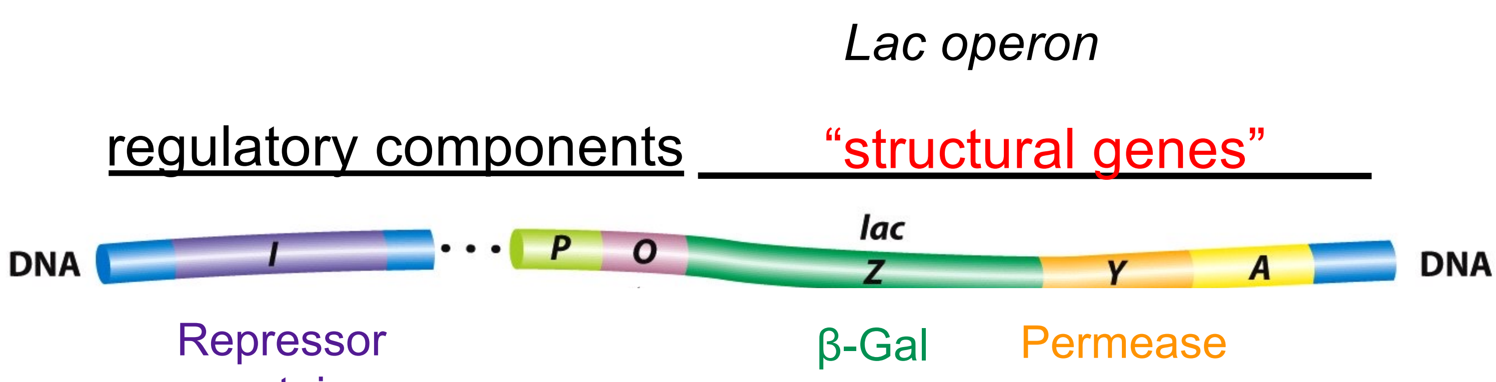


Fig.11-4

The *Lac* operon – a negative, inducible control system

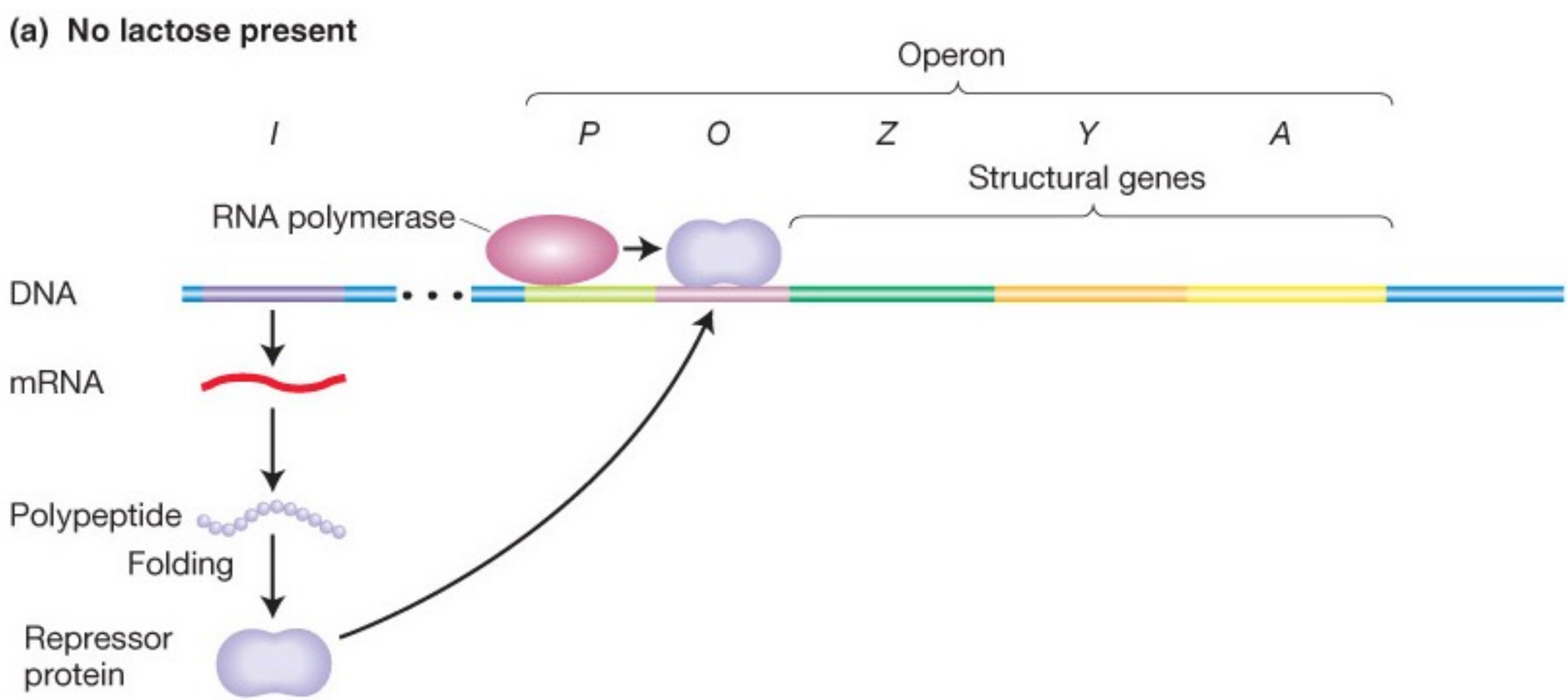


Fig.11-6a

The *Lac* operon – a negative, inducible control system

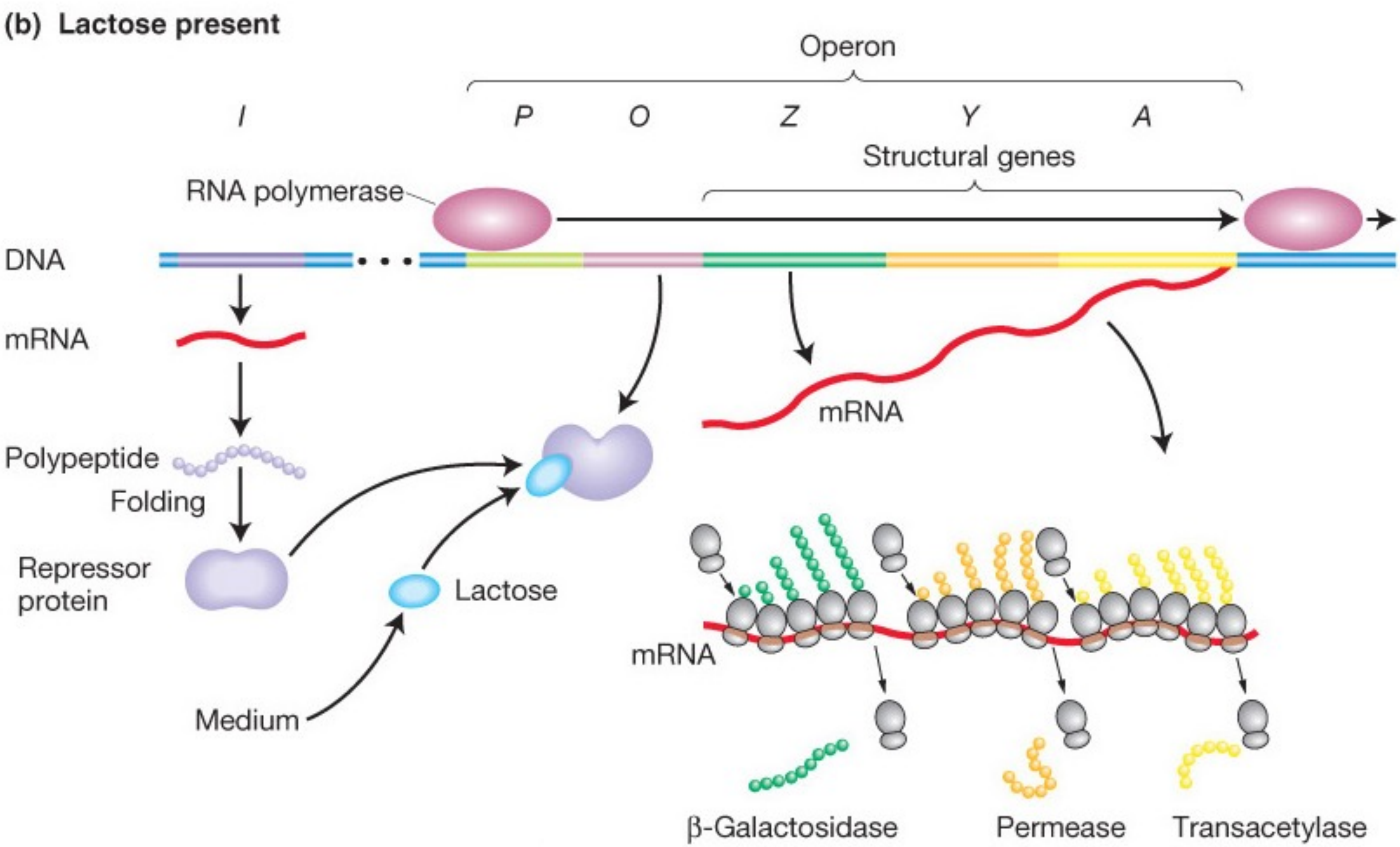


Fig.11-6b

Genetic analysis of the *lac* operon

- To uncover the regulatory mechanism of *lac* operon Jacob & Monod used a mutagenesis approach:

    treated *E. coli* with a chemical mutagen

    used biochemical assays to measure ß-galactosidase and/or Lac permease activity in presence and absence of a synthetic inducer (IPTG)

    selected mutant strains where one or both of these activities was altered.

Jacob and Monod's *lac* mutants

- <u>Three main classes of *lac* mutants</u>:

1. Structural gene mutations (affect function of just one enzyme – the other is **INDUCIBLE**)

	β-galactosidase		Permease	
Genotype	**no lactose**	**lactose**	**no lactose**	**lactose**
$Z^{+}\ Y^{+}$	-	+	-	+
$Z^{-}\ Y^{+}$	-	-	-	+

Bacteria grown in medium containing glycerol (no glucose)

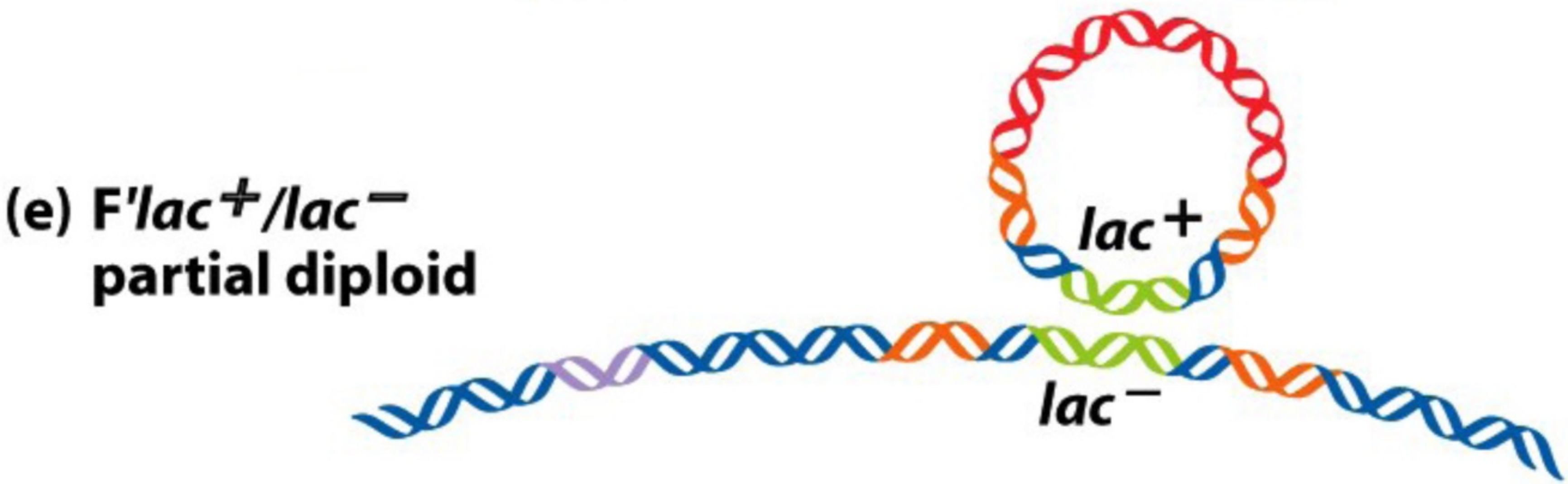
2. **UNINDUCIBLE** mutants – can't make LacZ <u>and</u> LacY in presence of inducer (IPTG)
3. **CONSTITUTIVE** mutants – make both LacZ and LacY even in absence of inducer

Use of partial diploids

- J+M used "partial diploid" bacteria harbouring a plasmid that contains a <u>second</u> copy of the *lac* operon

    allowed them to test dominance / recessiveness

    critical in determination of "cis" vs "trans"-acting factors



cis = only affects transcription of genes on same DNA molecule

trans = can also affect transcription of genes on the other DNA molecule

Think break…

Imagine two separate electrical circuits, each with its own battery, switch and light bulb. Switch 1 turns on and off light bulb 1, switch 2 turns on and off light bulb 2. The switches can only be activated by inserting a key, with the same key used for both switches.

In this setup – do the switches work in cis or in trans?

Do the batteries work in cis or in trans?

Does the key work in cis or in trans?

lac operon regulation: P mutations

	β-gal.		Perm.		
Genotype	no lactose	lactose	no lactose	lactose	
$P^+ Z^+ Y^+$	-	+	-	+	Inducible
$P^- Z^+ Y^+$	-	-	-	-	Uninducible
$P^+ Z^+ Y^+ / F' P^- Z^- Y^-$					Inducible
$P^- Z^+ Y^+ / F' P^+ Z^- Y^-$					Uninducible

Bacteria grown in medium containing glycerol (no glucose)

P^- affects inducibility of both β–Galactosidase and Permease - failure to "turn on"

P^- only affects transcription of genes physically attached to it on the same DNA molecule (i.e. chromosome or plasmid) – promoter is a **cis-acting element**.

lac operon regulation: P mutations

	β-gal.		Perm.		
Genotype	no lactose	lactose	no lactose	lactose	
$P^+ Z^+ Y^+$	-	+	-	+	Inducible
$P^- Z^+ Y^+$	-	-	-	-	Uninducible
$P^+ Z^+ Y^+$ / F' $P^- Z^- Y^-$	-	+	-	+	Inducible
$P^- Z^+ Y^+$ / F' $P^+ Z^- Y^-$	-	-	-	-	Uninducible

Bacteria grown in medium containing glycerol (no glucose)

P^- affects inducibility of both β–Galactosidase and Permease - failure to "turn on"

P^- only affects transcription of genes physically attached to it on the same DNA molecule (i.e. chromosome or plasmid) – promoter is a **cis-acting element**.

lac operon regulation: O mutations

	β-gal.		Perm.		
Genotype	no lactose	lactose	no lactose	lactose	
$O^+ Z^+ Y^+$	-	+	-	+	Inducible
$O^C Z^+ Y^+$	+	+	+	+	Constitutive
$O^+ Z^- Y^- / F' O^C Z^+ Y^+$					Constitutive
$O^+ Z^+ Y^+ / F' O^C Z^- Y^-$					Inducible

O^C (= O^-): constitutive… both β–Galactosidase and Permease are constitutively active - failure to "keep off"

similar to P, O only affects transcription of genes physically attached to it on the same DNA molecule – operator is a **cis-acting element**.

lac operon regulation: O mutations

	β-gal.		Perm.		
Genotype	no lactose	lactose	no lactose	lactose	
$O^+ Z^+ Y^+$	-	+	-	+	Inducible
$O^C Z^+ Y^+$	+	+	+	+	Constitutive
$O^+ Z^- Y^- / F' O^C Z^+ Y^+$	+	+	+	+	Constitutive
$O^+ Z^+ Y^+ / F' O^C Z^- Y^-$	-	+	-	+	Inducible

O^C (= O^-): constitutive… both β–Galactosidase and Permease are constitutively active - failure to “keep off”

similar to P, O only affects transcription of genes physically attached to it on the same DNA molecule – operator is a **cis-acting element**.

Promoters and Operators are cis-acting DNA sequences

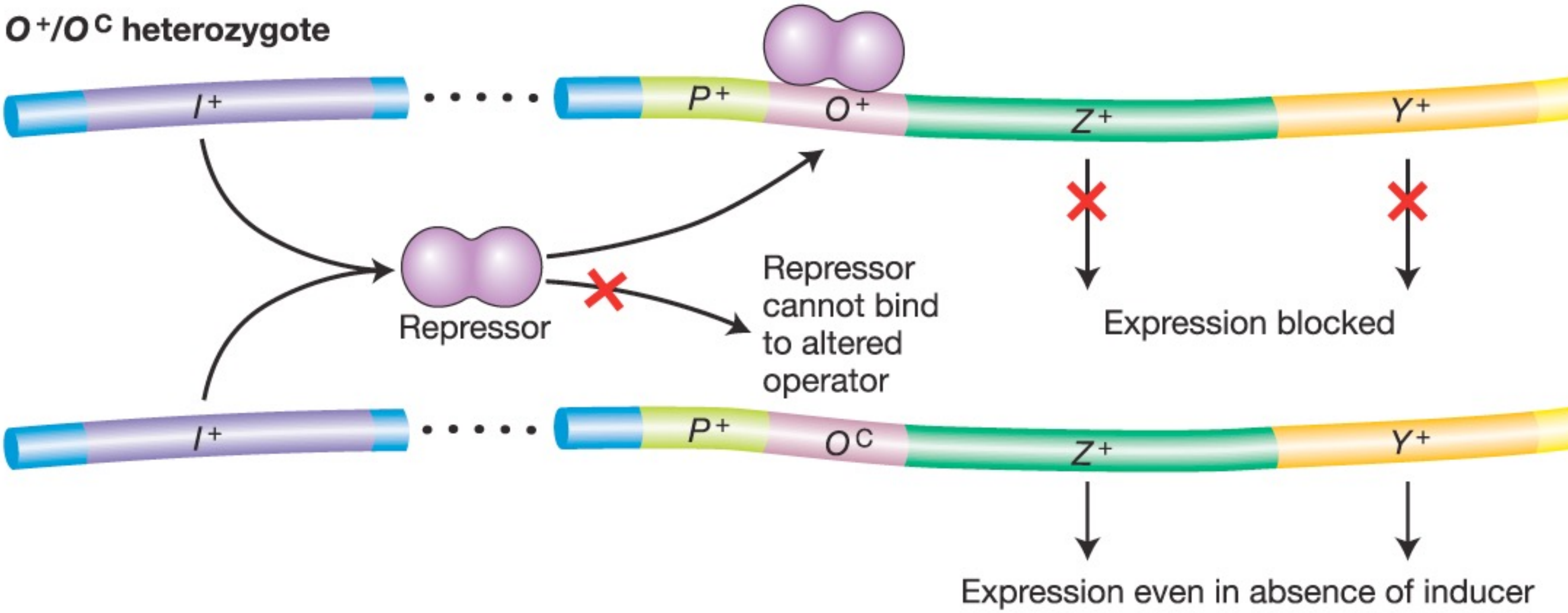


Fig.11-8

lac operon regulation: I^- mutations

	β-gal.		Perm.		
Genotype	no lactose	lactose	no lactose	lactose	
$I^+ Z^+ Y^+$	-	+	-	+	Inducible
$I^- Z^+ Y^+$	+	+	+	+	Constitutive
$I^+ Z^- Y^- / F' I^- Z^+ Y^+$					Inducible

Bacteria grown in medium containing glycerol (no glucose)

I^- : similar to O, since both β–Galactosidase and Permease are constitutively active

<u>However</u>, *I* doesn't have to be on the same DNA molecule – can **act in trans**…
diffusible factor

lac operon regulation: I^- mutations

	β-gal.		Perm.		
Genotype	**no lactose**	**lactose**	**no lactose**	**lactose**	
$I^+ Z^+ Y^+$	-	+	-	+	Inducible
$I^- Z^+ Y^+$	+	+	+	+	Constitutive
$I^+ Z^- Y^- / F' I^- Z^+ Y^+$	-	+	-	+	Inducible

Bacteria grown in medium containing glycerol (no glucose)

I^- : similar to O, since both β–Galactosidase and Permease are constitutively active

<u>However</u>, *I* doesn't have to be on the same DNA molecule – can **act in trans**…
diffusible factor

Conclusion: LacI gene must encode a diffusible, repressive protein (*trans*-acting)

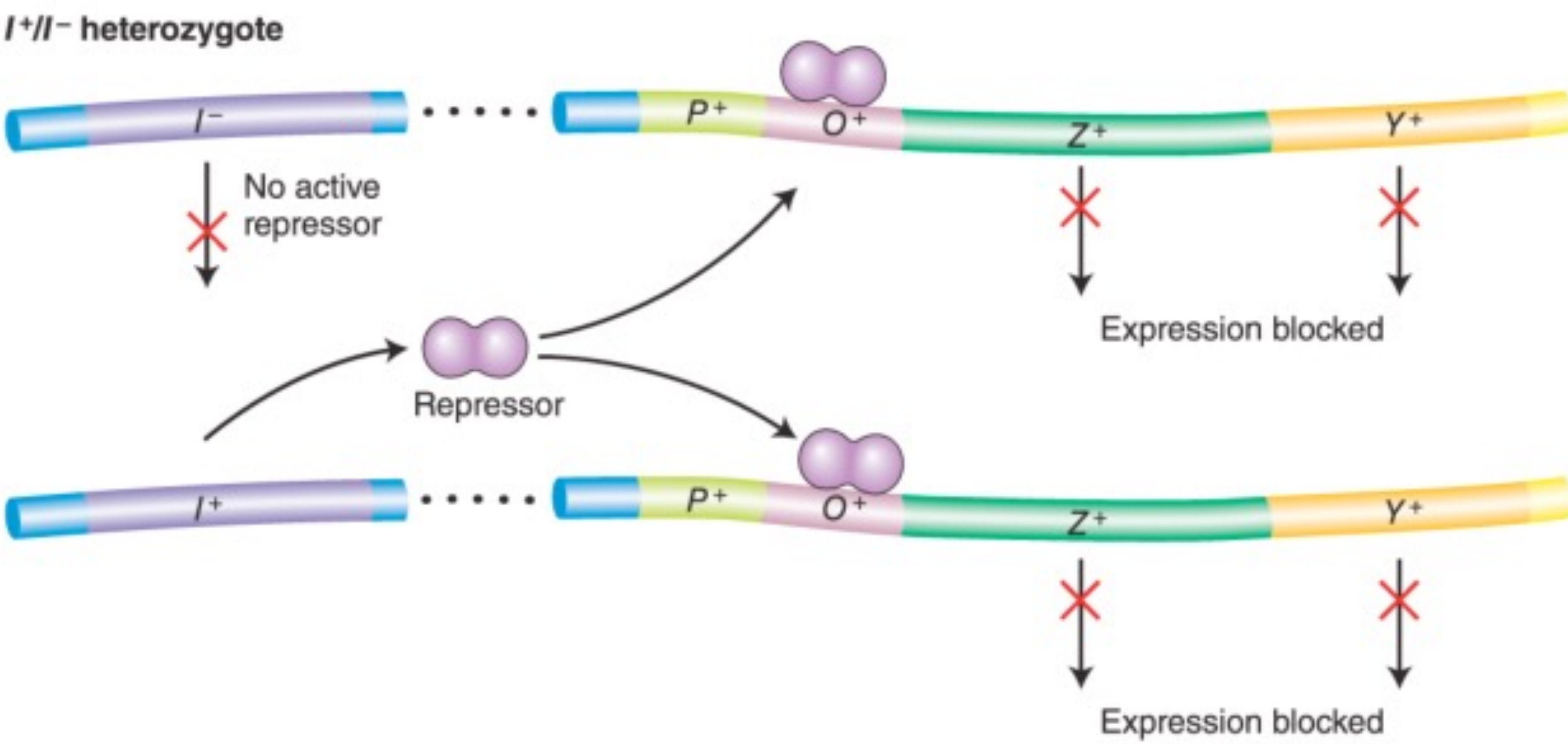


Fig.11-9

lac operon regulation: I^S mutations, "super-repressor"

	β-gal.		Perm.		
Genotype	Lac-	Lac+	Lac-	Lac+	
$I^+ Z^+ Y^+$	-	+	-	+	Inducible
$I^S Z^+ Y^+$	-	-	-	-	Uninducible
$I^S Z^+ Y^+ / F' I^+ Z^+ Y^+$					Uninducible

Bacteria grown in medium containing glycerol (no glucose)

I^S is different from I^-; it cannot be complemented by I^+ (i.e. it's dominant to I^+)

lac operon regulation: I^S mutations, "super-repressor"

	β-gal.		Perm.		
Genotype	Lac-	Lac+	Lac-	Lac+	
$I^+ Z^+ Y^+$	-	+	-	+	Inducible
$I^S Z^+ Y^+$	-	-	-	-	Uninducible
$I^S Z^+ Y^+ / F' I^+ Z^+ Y^+$	-	-	-	-	Uninducible

Bacteria grown in medium containing glycerol (no glucose)

I^S is different from I^-; it cannot be complemented by I^+ (i.e. it's dominant to I^+)

I^S mutations affect the allosteric site of LacI

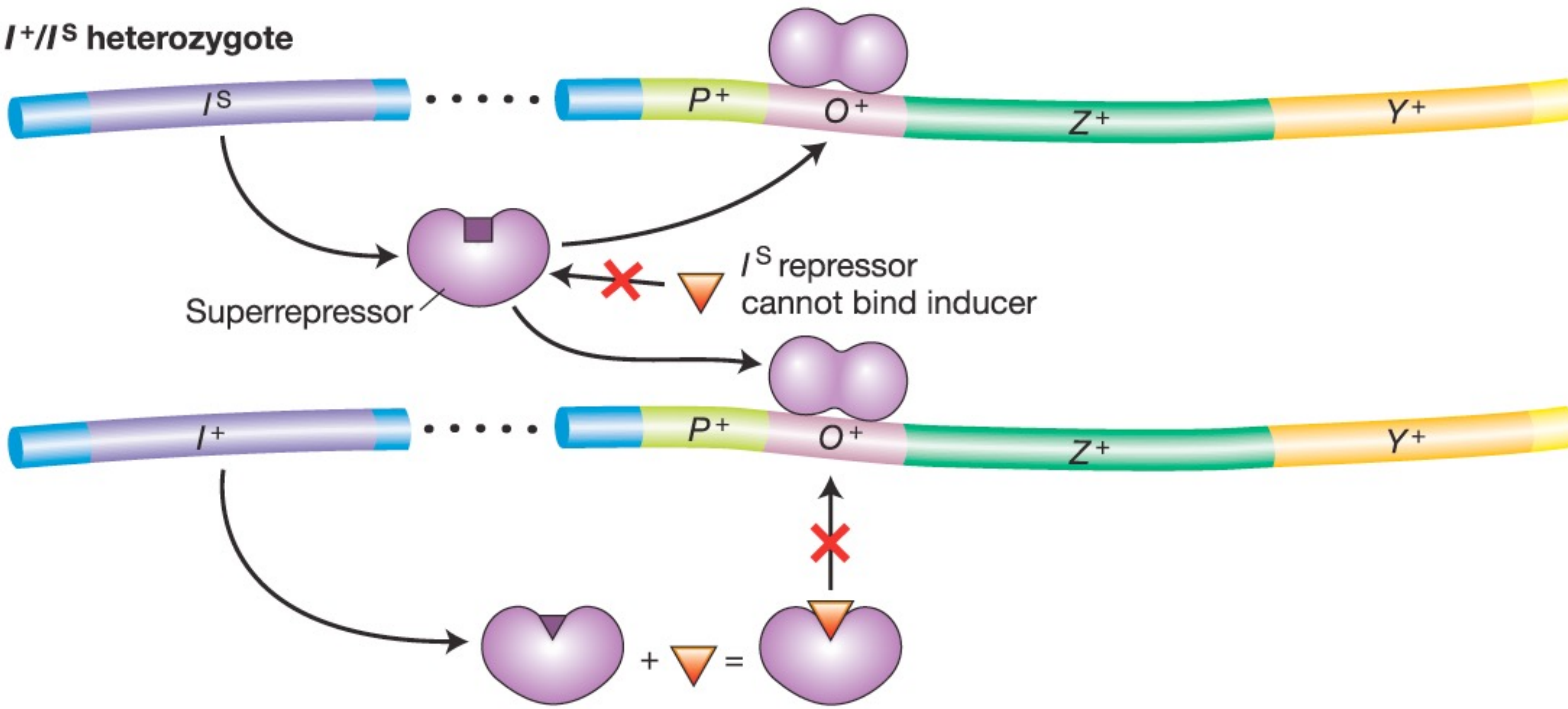


Fig.11-10

lac operon sample question

What would be the phenotype for the following *lac* operon partial diploid?

1. $I^+ P^+ O^+ Z^- Y^+ / F' \, I^+ P^- O^C Z^+ Y^-$

   a. Z is constitutive, Y is inducible
   b. Z is inducible, Y is inducible
   c. Z is uninducible, Y is inducible
   d. Z is constitutive, Y Is uninducible

It’s more complicated…

If you REALLY love doughnuts, but are only so-so about chocolate chip cookies…

Obviously, you’ll eat doughnuts when you need a snack and they’re just sitting in the kitchen.

If there are no doughnuts available, you’ll eat the cookies – better than nothing.

But if there are doughnuts AND cookies in the kitchen you’ll probably start with the doughnuts, and eat the cookies only once the dough-nuts are gone, no?

The *lac* operon is also under positive regulation

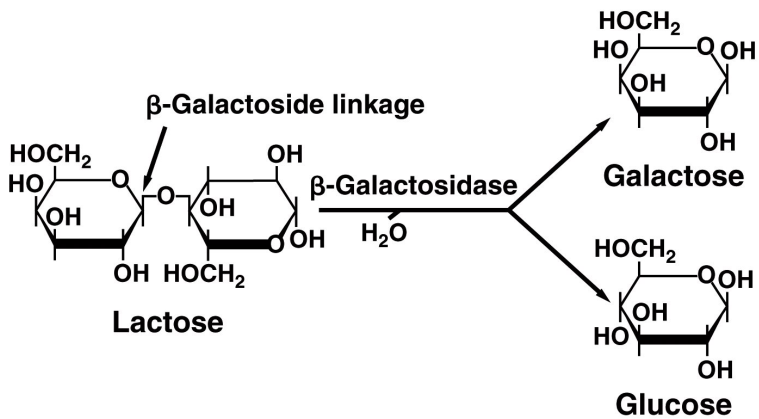


Fig.11-5

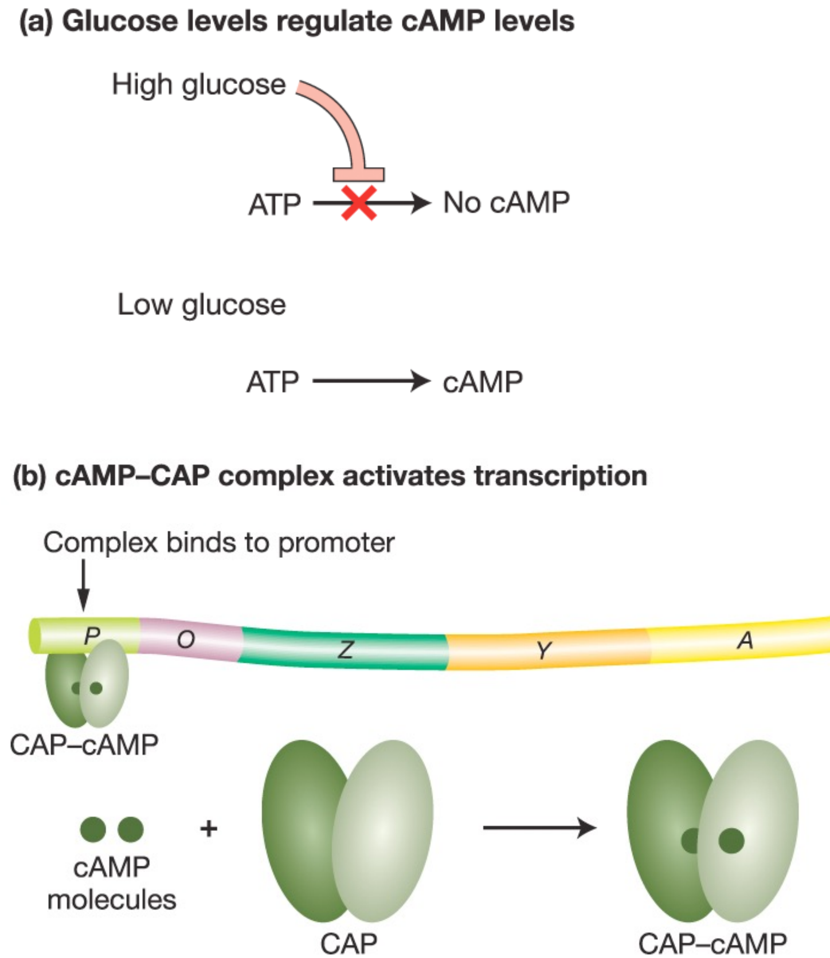
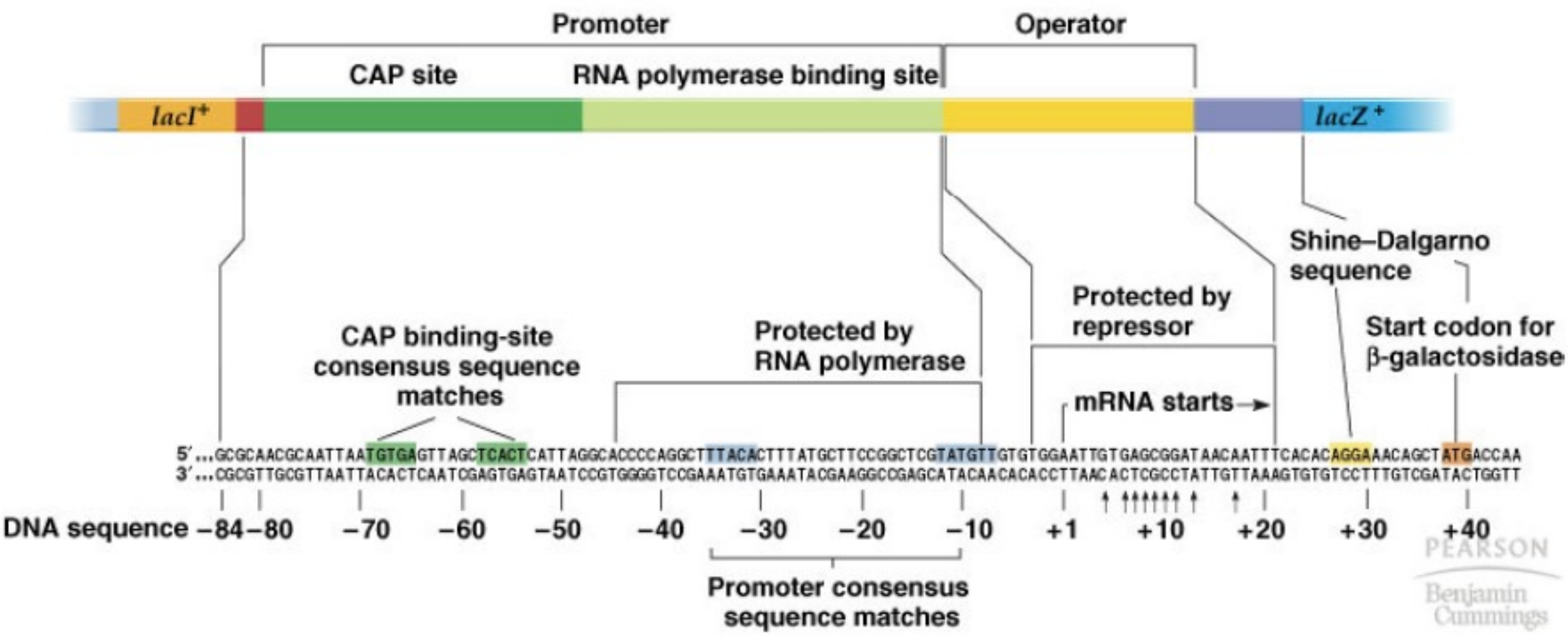


Fig.11-13

- **CAP**: <u>Catabolite Activator Protein</u> coded by the *crp* gene
- CAP-cAMP binds P just upstream of RNApol
- facilitates RNApol binding to P to increase transcription

Organization of the *lac* control region



Integrating positive and negative control of the *lac* operon

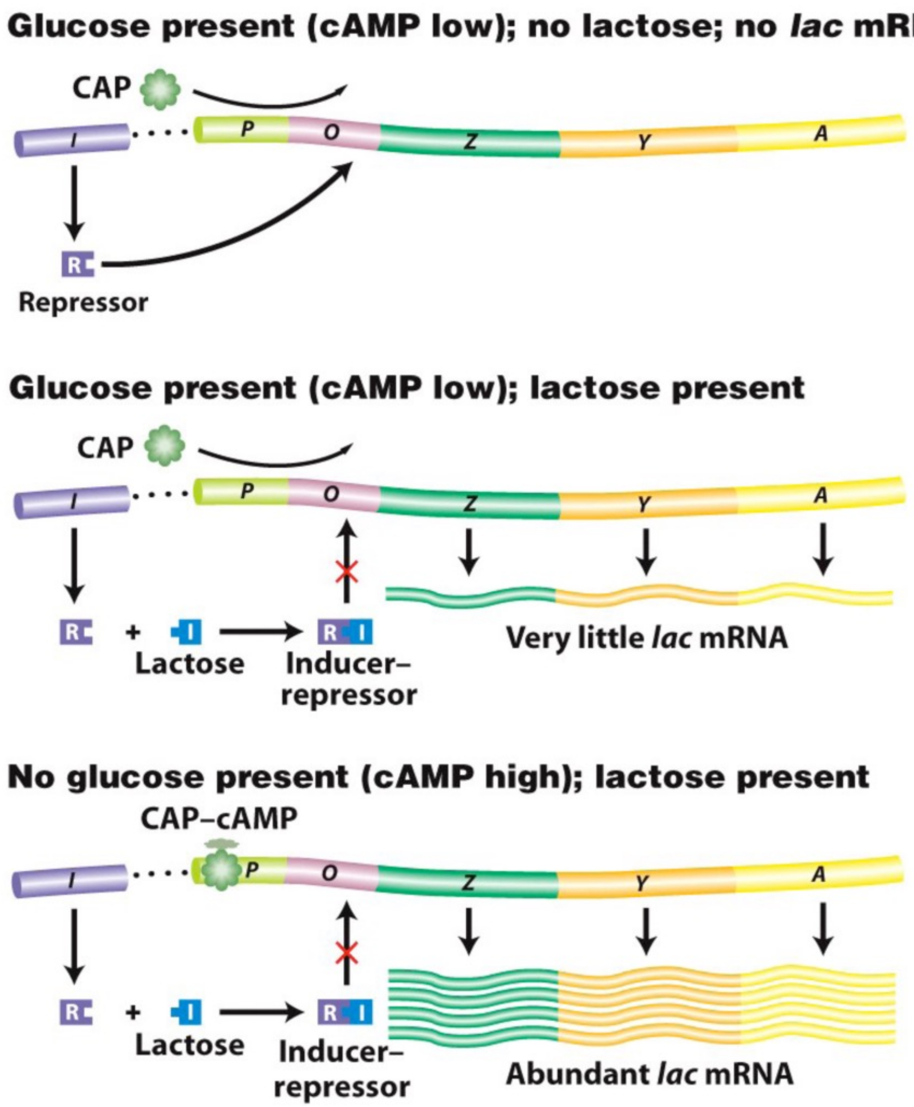


Fig.11-17